Solar Energy Materials and Solar Cells 192 (2019) 72–80

Contents lists available at ScienceDirect

Solar Energy Materials and Solar Cells

journal homepage: www.elsevier.com/locate/solmat

Enhanced thermal conductivity of microencapsulated phase change materials based on graphene oxide and carbon nanotube hybrid filler

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ARTICLE INFO

Keywords:
Microencapsulated phase change material
Graphene oxide
Carbon nanotube
Hybrid filler
Thermal conductivity
Photo-thermal conversion performance

ABSTRACT

Large energy storage capacity and high thermal charging/discharging rate are crucial for microencapsulated phase change materials (MEPCM) in thermal energy storage application. The microcapsules with dodecanol core and melamine-formaldehyde (MF) resin shell modified by graphene oxide (GO) and carbon nanotube (CNT) hybrid filler were prepared via in-situ polymerization. The effect of the combination of GO and CNT on morphology and thermal properties of the microcapsules was investigated, and the mechanism for heat transfer enhancement was further studied from both macroscopic and microscopic views. The results indicated that the addition of GO and CNTs largely enhanced the thermal conductivity of the microcapsules, and the GO-CNT hybrid filler was superior to individual GO or CNT. Particularly, with a hybrid filler loading of 0.6 wt%, the thermal conductivity of the microcapsule was improved by 195% with the latent heat slightly decreased. The prepared microcapsules were dispersed into water to form a latent functional thermal fluid, and its photo-thermal conversion performance was studied. The excellent thermal properties and photo-thermal conversion performance made the microcapsule dispersed slurry a potential fluid in direct absorption solar collectors.

1. Introduction

Non-renewable fossil fuels and the growing shortage are a major problem facing our country and the world. Solar thermal utilization is an effective way to utilize solar radiation and alleviate the energy crisis and the environmental problems caused by the consumption of fossil energy. Direct absorption solar collector (DASC) is a new kind of solar collectors, which is the key component of the solar thermal utilization systems. The thermophysical properties and photo-thermal conversion performance of its working fluid directly determine the efficiency of the direct absorption solar collector. Latent heat storage fluid, with three excellent characteristics of heat collection, heat storage, and heat transfer, is a recently developed heat transfer fluid, formed by dispersing PCM or MEPCM into base fluid. Latent heat storage makes use of the endothermic and exothermic heat of phase change materials when they melt and solidify. PCMs are liable to flow and react with the outside environment when directly used [1–3], and the microencapsulation technology is one of the best solutions to solve these problems. Nevertheless, the most widely used coating shells are organic, for example, PMMA [4–7], MF [8], PS [9,10], PU [11] and epoxy resin [12], which reduce the energy charging and discharging rates of the microencapsulated phase change materials (MEPCM) due to their low thermal conductivity.

To improve the heat storage efficiency of MEPCM, many high conductive fillers or inserts are employed to enhance the thermal conductivity of phase change materials [13–16], such as Al_2O_3 [17], Si_3N_4 [18,19], graphite [20], expanded graphite [21,22] et al. Graphene is generally used as the filler to improve the thermal conductivity based on its high thermal conductivity of about 5300 W/m K [23]. Several researches have used graphene as the filler to improve the thermal conductivity of PCMs [24–26], but it has only recently been employed in MEPCM [27–31]. Wang et al. [32] mixed n-eicosane/silica microcapsules with thermal reduced rGO (rGO-HT) and $NaBH_4$ reduced rGO (rGO-$NaBH_4$). With the addition of 1 wt%, rGO-HT resulted in a 193% increase in thermal conductivity and a 15% decrease in the latent heat, while rGO-$NaBH_4$ increased the thermal conductivity by 83% and decreased the latent heat by 6%. This is due to the difference of surface morphology and functional groups between rGO-HT and rGO-$NaBH_4$ sheets. Chen et al. [33] synthesized the dodecanol/poly(melamine-formaldehyde) microcapsule modified by graphene oxide. With 4 wt% dosing, the thermal conductivity was enhanced by 66.29%.

Carbon nanotube (CNT) has the advantages of high aspect ratio, outstanding thermal conductivity and low density, thus it is a promising thermal-conductive filler [34–37]. CNTs were simultaneously

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https://doi.org/10.1016/j.solmat.2018.12.014
Received 14 September 2018; Received in revised form 6 December 2018; Accepted 7 December 2018
Available online 18 December 2018

introduced into the PCM core and the melamine–formaldehyde pre-polymer shell material of the microcapsule, and the thermal conductivity was enhanced by 25% with 1.67 wt% CNTs [38]. Huang et al. [39] prepared the carbon nanotube/MF double walled microcapsules via layer-by-layer self-assembly technique. With 4 assembled CNT layers, the thermal conductivity was enhanced by 57.89% at most. To reduce the aggregation and improve the dispersion of CNTs caused by the inactive surface of CNTs, Sun et al. [40] modified CNTs with O_2-plasma, and added O_2-plasma-modified CNTs into the emulsion system or the polymer system. When 0.2 g of modified CNTs were added to the polymer system, the thermal conductivity of the microcapsule was improved by 225%.

Recent studies indicated that 1D CNTs can integrate with 2D graphene to create 3D nanostructure, and the hybridization of CNTs and graphene outperformed their corresponding CNTs or graphene in thermal property [41–43]. Some studies have combined graphene with CNTs to improve the thermal conductivity of many kinds of materials, for example paraffin [44,45], PS [46,47], epoxy resin [48,49], PVDF [50,51], etc. It has been demonstrated that there is a synergistic effect between graphene and CNTs in improving the thermal conductivity and the performance of CNT and graphene hybrid filler is superior to the individual graphene or CNTs [52]. Carbon nanomaterials including GO and CNT were also added into water to obtain nanofluid for DASCs due to their optical absorption property [53,54]. To the best of our knowledge, there are few studies applying graphene and CNT hybrid fillers into the MEPCM, especially investigating the mechanism for heat transfer enhancement with GO-CNT hybrid filler in MEPCM, and rare researches have investigated the photo-thermal conversion performance of the latent heat storage fluid containing the microcapsules with GO and CNT fillers. In the current work, we added GO and CNTs into the dodecanol/melamine-formaldehyde resin microcapsules and investigated the effect of the combination of GO and CNT on the thermal energy storage capacity and storage efficiency of the microcapsules. Moreover, we dispersed the microcapsules modified by GO and CNT into water to form latent functional thermal fluids, and studied their photo-thermal conversion performance.

2. Materials and experiments

2.1. Materials

Graphite and CNT with carboxyl groups (outer diameter of more than 50 nm, length of about 0.5–2 μm) were provided by Nanjing XF Nano Materials Tech Co. Ltd. The Dodecanol was obtained from Aladdin Reagent (Shanghai) Co., Ltd. Styrene maleic acid copolymer sodium salt (SMA1000HNa) was provided by Sartomer, USA. Other reagents were bought from Guangdong Chemical Reagent Corporation and all the reagents were used as received without further purification.

2.2. Synthesis of GO aqueous dispersion

The method of preparing GO aqueous dispersion was the same as what was presented in our previous research [55]. Briefly, 2 g graphite, 1 g sodium nitrate and 46 mL concentrated sulfuric acid were added into a three-neck flask under an ice bath and stirred for 30 min. Afterwards, 6 g potassium permanganate was slowly added into the mixed liquid and agitated for another 2 h. Subsequently, the flask was moved to an oil bath of 35 °C and agitated for 30 min. Next, 92 mL distilled water was slowly added and the temperature was increased to 98 °C. After 15 min of stirring, 15 mL H_2O_2 (30%) was added and then 140 mL distilled water was added to dilute the mixture. Finally, the reaction fluid was washed by distilled water until the solution was neutral and exfoliated by ultrasonication.

2.3. Synthesis of MEPCM modified by GO and CNT

The MEPCM samples were prepared by in-situ polymerization. 10 g melamine and 21.5 g formaldehyde were added into a 250 mL three-neck flask and heated to 70 °C until the mixture became transparent. Meanwhile, GO aqueous dispersion and CNT powder were ultrasonically dispersed for 20 min. Then the dispersion liquid was added to the mixture and the pH was adjusted to 8–9. The reaction liquid was kept at 70 °C for 3 h to obtain the shell pre-polymer. 25 g dodecanol, 30 g water and 2.5 g SMA1000HNa were mixed and vigorously stirred for 1 h to get the core pre-emulsion, and then the emulsion was adjusted to a pH of 3–4. The shell pre-polymer was dropped into the core emulsion and stirred for another 3 h. Finally, the reaction product was filtered and washed, then dried at 50 °C.

The latent functional thermal fluids were prepared by dispersing microcapsule powders into deionized water with the aid of sodium dodecyl sulfate (SDS) and stirred for 20 min.

2.4. Characterization and measurement

The morphology of the MEPCM samples and the fractured surface of the polymer shells were observed by Scanning electron microscope (SEM, FEI Nova NanoSEM 430, Holland). 4 g shell pre-polymer was added to the vessel and then dried to obtain a thin film. The fractured surface of the polymer shells was obtained by breaking the dried film. The MEPCM powder samples were directly sprinkled on conductive adhesive and sprayed with gold. The chemical composition of the microcapsules was analyzed by Fourier Transform Infrared Spectrometer (Bruker Dalton, Tensor II) at room temperature. The microcapsule powder was mixed with KBr pellets and the wave numbers ranged in 4000–500 cm^{-1}. Differential scanning calorimeter (DSC, 200 F3, NETZSCH, Germany) was used to analyze the phase change properties of MEPCMs, and the testing temperature was in the range of −30–60 °C with the heating/cooling rate of 10 °C/min under nitrogen atmosphere. The mass weight of every sample was about 5 mg. Hot Disk (TPS2500, Sweden) was employed to measure the thermal conductivity of microcapsules at room temperature. 10 g MEPCM sample was added into a cylindrical mold and compacted to obtain two flat surfaces, and the probe was put between them. Thermal stability of MEPCM was tested by thermos-gravimetric analysis (TG 209F, Netzsch). 8–10 mg for every sample was heated from 30 °C to 600 °C at the heating rate of 10 °C/min under nitrogen atmosphere. In order to investigate the thermal cycle stability, the MEPCMs experienced several heating/cooling cycles between −30 °C and 50 °C, and the latent heat was measured by DSC after several cycles. The Zeta potential of the suspensions was examined by Malvern instrument (Nano-ZS90, England) at 30–80 °C with the interval of 10 °C. The light absorption property of the microencapsulated phase change slurries was measured by the UV–vis spectroscopy (Agilent cary60, USA). The photo-thermal conversion performance of the slurries was investigated by the instrument as shown in Fig. 1. The obtained suspension was put into the quartz baker and the photo-thermal conversion occurred when the irradiation started. The temperature of the suspension was measured by the thermocouple (J-type) and the data were recorded by Data collector (Agilent 34972A, Malaysia).

3. Results and discussion

3.1. Morphology of the microcapsules

The surface morphology of the microcapsules and the fractured surface of the microcapsule shells were detected by SEM. As shown in Fig. 2, all the microcapsules are regular spherical structure with smooth surfaces, and the diameters range from 300 to 800 nm. However, some exposed tubes can be obviously seen in Fig. 2(c) and (d),

which indicates that CNTs exist in the microcapsules with CNT and GO-CNT. CNT has a large aspect ratio and its length is longer than the

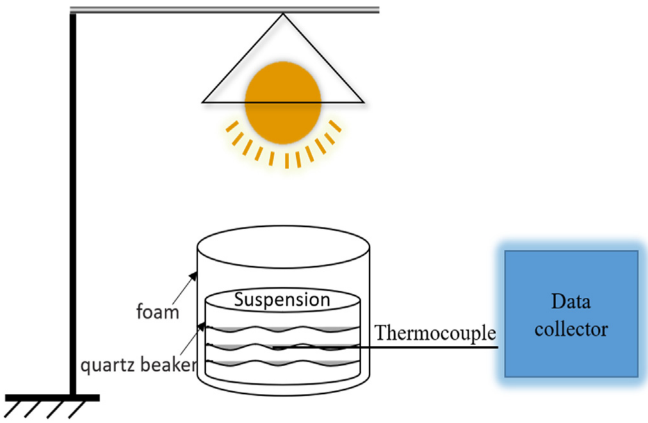

Fig. 1. Experimental setup for evaluating the photo-thermal conversion performance.

grain size of microcapsules, thus CNTs may not be entirely packed into the microcapsule and part of the nanotubes extend out of the microcapsule. Even so, the morphology of MEPCM with GO or CNT is similar to that of the MEPCM without any filler, which indicates that the addition of GO or CNTs into the shell has little effect on the overall morphology of the microcapsules.

Fig. 3 are the fractured surface images of melamine-formaldehyde (MF) resin shell with GO or CNTs as fillers, which can reflect the dispersion state of fillers in the shell and the interfacial interaction between them. Fig. 3(a) and (b) are the SEM images of GO and CNT. GO is a two-dimensional material with laminar structure and CNT is a one-dimensional material with banded structure. The two kinds of materials can intertwine with each other to form a 3-D network structure, which can be observed in the fractured surface image of melamine shell with GO and CNT simultaneously, as shown in Fig. 3(h). Pure melamine presents a smooth fractured surface as shown in Fig. 3(c), while the fractured surfaces of resin with fillers are rougher. The SEM image of melamine resin with GO is presented in Fig. 3(d) and uniformly dispersed wrinkles can be seen, which is the characteristic of GO. Fig. 3(e) shows the fractured surface of melamine resin with CNTs, in which, several holes formed by the shedding of CNTs from resins and the naked nanotubes not soaked by resins can be obviously seen. This demonstrates the poor interface contact between CNTs and the melamine resin [56]. As displayed in Fig. 3(f) and (g), both GO and CNTs can be obviously seen in the fractured surface of MF/GO-CNTs, and some holes are also present. The results suggest that GO has a relatively good interface compatibility with the melamine shell and the presence of CNT promotes the dispersion of GO in the shell, but the interfacial interaction between CNT and the melamine resin is poor.

3.2. Chemical composition of the microcapsules

The FTIR spectra of the microcapsules with GO or CNT were shown in Fig. 4. In the microcapsule without any filler, the absorption peaks at 2924 and 2843 cm^{-1} were assigned to the stretching vibration bands of C–H in $–CH_3$ and $–CH_2–$, and the absorption peaks at 1469 and 1343 cm^{-1} were the bending vibration bands of C–H in $–CH_3$ and $–CH_2–$. The absorption peaks at 1048 cm^{-1} corresponded to the stretching vibration of C–OH. They were all characteristic peaks of n-dodecanol. The peaks at 1559 and 815 cm^{-1} were the stretching and bending bands of the triazine ring. The peak at 3346 cm^{-1} was the overlap of the stretching bands of N–H and O–H, and the peak at 1512 cm^{-1} was the bending band of N–H. The absorption peak at 1163 cm^{-1} was the stretching bands of C–O–C. The above peaks were the characteristic peak of MF resin. Notably, the spectra of the microcapsules with GO, CNT or GO-CNT hybrid filler were similar with that

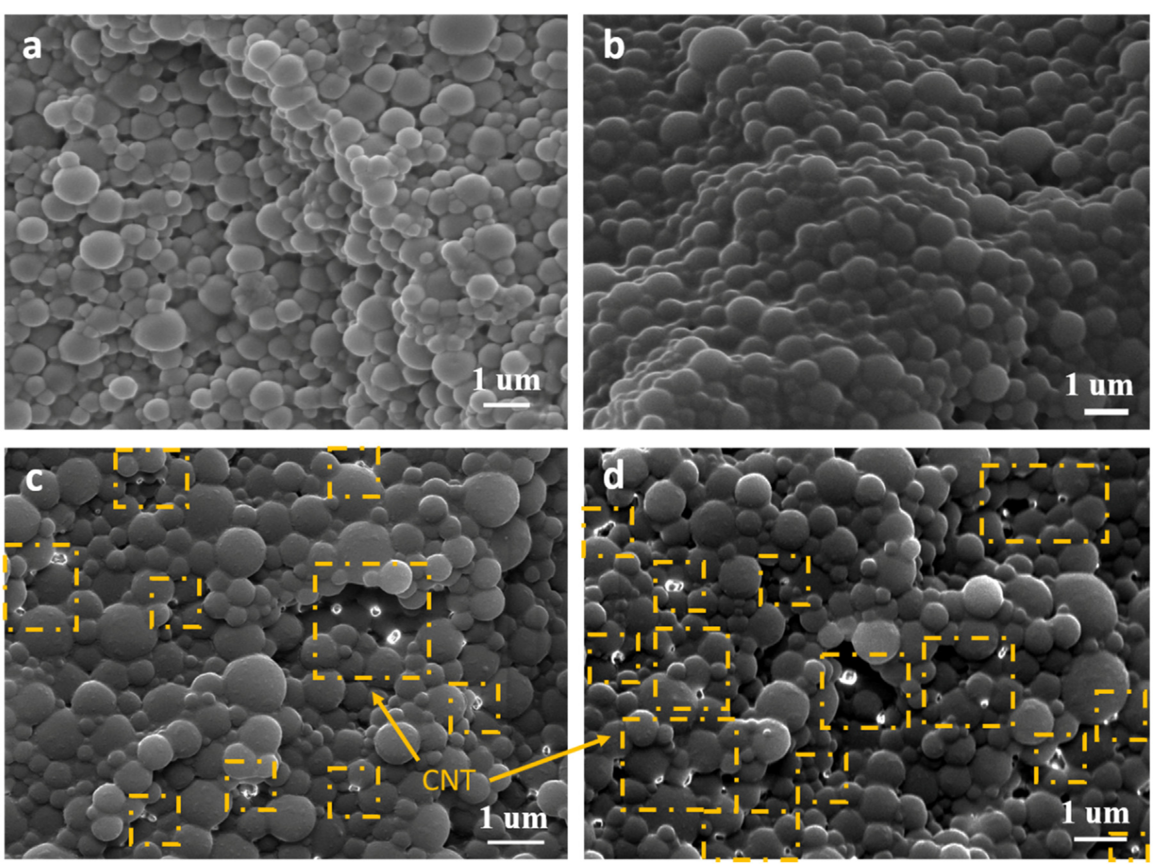

Fig. 2. SEM images of (a) MEPCM, (b) MEPCM/GO, (c) MEPCM/CNT, (d) MEPCM/GO-CNT.

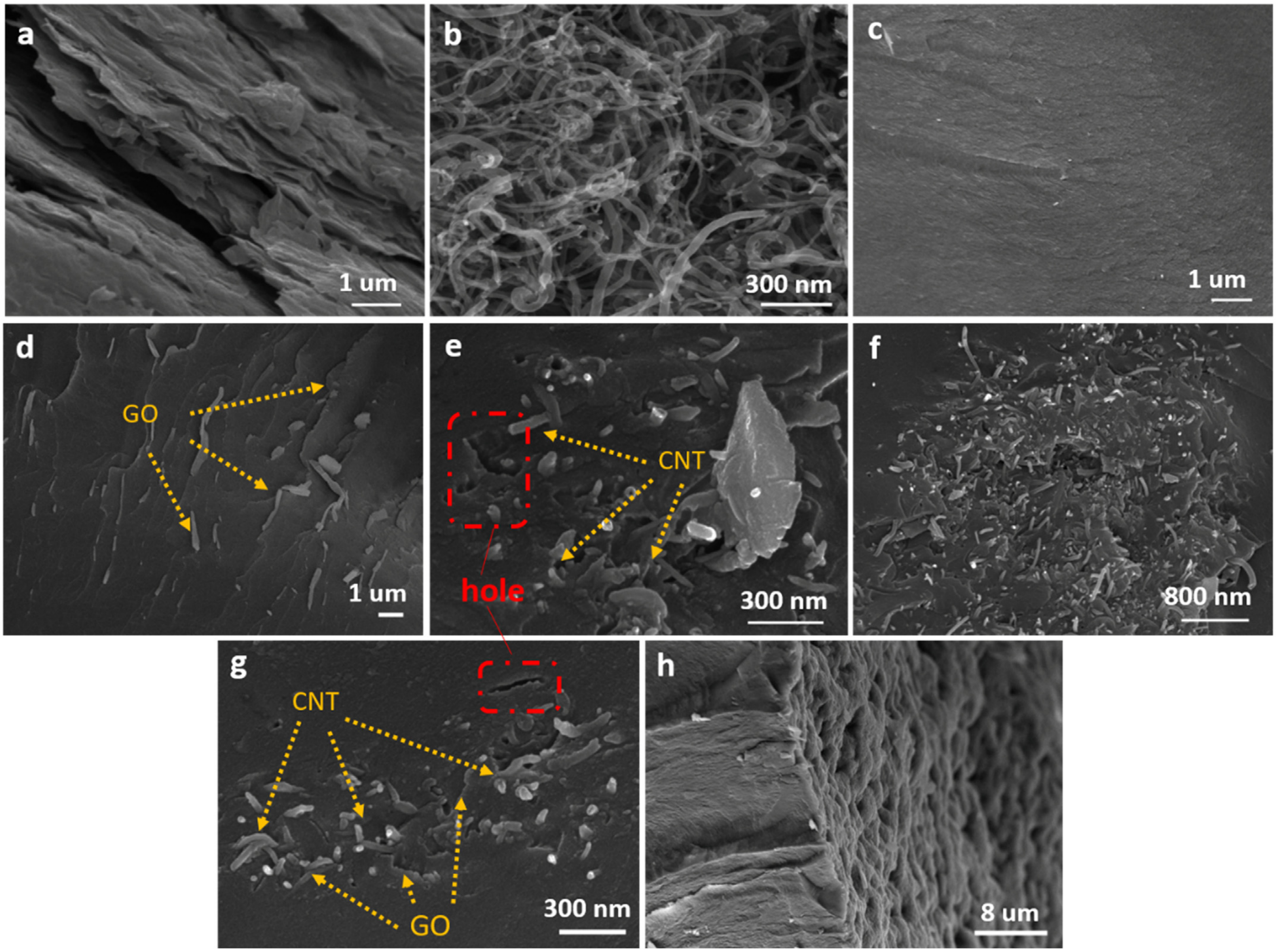

Fig. 3. SEM images of (a) GO, (b) CNT, fracture morphology of (c) MF, (d) MF/GO, (e) MF/CNT, and (f–h) MF/GO-CNT.

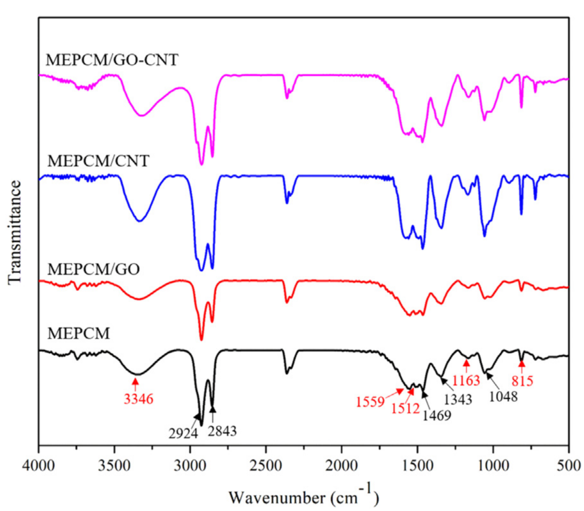

Fig. 4. FTIR spectrums of the microcapsules without or with GO or CNT.

of the microcapsule without any filler, which indicated that the addition of GO or CNT had little influence on the chemical structure of the microcapsules.

3.3. Thermal conductivity of the microcapsules

It is well known to all that the interface interaction between the filler and the matrix and the filler dispersion state in the matrix are of great importance to the thermal conductivity [57]. The former is related with the interfacial resistance, while the latter is connected with the formation of conductive path. From the discussion about Fig. 3, there is a good interface interaction between GO and the polymer shell, however, the interface interaction between CNT and the shell is poor. To better observe the dispersion of different fillers in the polymer shell, the pre-polymer of the shell with brown GO or black CNTs was poured into a cylinder mold and then cured and dried. The macroscopic morphology of the cured composites is shown in Fig. 5. The top and bottom surfaces of pure melamine resin are both milky white. The melamine resin with GO has a completely brown bottom surface but a white with brown top surface. CNT-modified melamine resin has a black with white bottom surface and an entirely white top surface. This indicates that GO has a good dispersion in the polymer shell while the dispersion of CNT is poor and most CNTs are deposited in the bottom. Both the bottom and top surfaces of melamine resin with GO and CNT are uniformly black. In addition, the cured cake of MEPCM/GO-CNT is broken and all the obtained fractured surfaces are black throughout sample thickness. This indicates that GO-CNT hybrid filler is uniformly dispersed in the whole polymer shell, which is due to the reason that the presence of CNT promotes the dispersion of GO in the shell and the formed filler network homogeneously disperses in the shell [50].

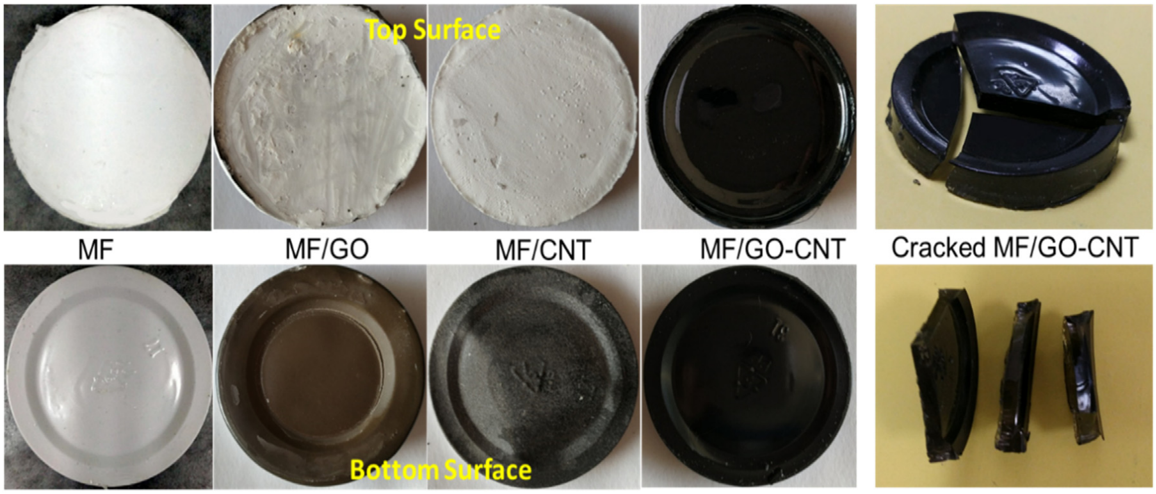

Fig. 5. Dispersion state of the shell polymer with GO and CNT.

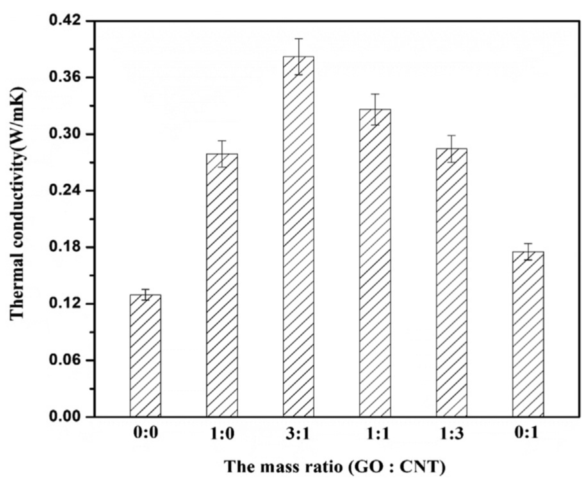

Fig. 6. Effect of the mass ratio of GO and CNT on the thermal conductivity of MEPCM/GO-CNT.

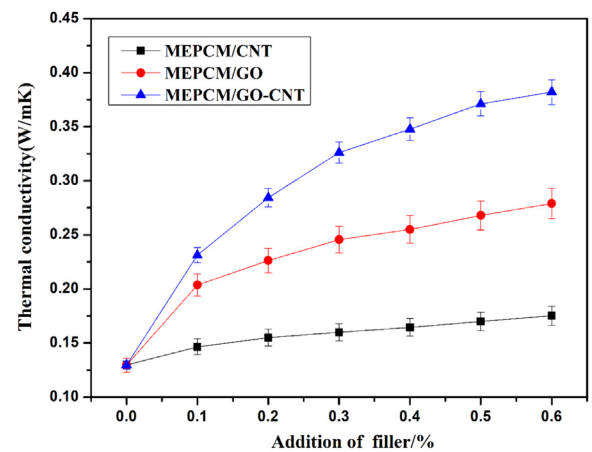

Fig. 7. Thermal conductivity of the microcapsules with different fillers.

From the discussion above, the interface interaction of GO with the shell is better than that of CNT with the shell. In order to investigate the effect of the addition of GO and CNT on the thermal conductivity of the microcapsule, we mixed GO and CNT. Fig. 6 shows the thermal conductivity of MEPCM/GO-CNT at different mass ratio of GO and CNT, with the total addition of the hybrid filler fixed at 0.6 wt%. The results indicate that the thermal conductivity of the microcapsule with only GO is higher than that of the microcapsule with single CNT. In addition, the thermal conductivity of the microcapsule is sharply increased when a small amount of GO is replaced by CNT. When the mass ratio of GO and CNT is 3:1, namely 0.45 wt% GO and 0.15 wt% CNTs in the microcapsule, the maximum thermal conductivity is obtained. This phenomenon suggests that the GO and CNT hybrid filler is superior to single GO or CNT in enhancement of the thermal conductivity of the microcapsules.

To further study the synergistic effect of GO and CNT hybrid fillers, the thermal conductivity of MEPCM with different fillers is investigated as shown in Fig. 7. With the increase of the filler content, the thermal conductivity of all the three microcapsules increases with different extents. The thermal conductivity of the microcapsule without any fillers is 0.1296 W/mK. GO alone can improve the thermal conductivity of the microcapsules to 0.2790 W/mK with the addition of 0.6 wt%. The thermal conductivity of the microcapsule rises slowly with only CNT. With 0.6 wt% CNT, the thermal conductivity of MEPCM/CNT is 0.1752 W/mK, only enhanced by 35.2%. This is due to the poor dispersion state of CNT in the polymer shell and the weak interface interaction between CNT and the shell, leading to severe phonon scattering and enhanced interface thermal resistance. With the weight ratio of GO and CNT fixed at 3:1, the thermal conductivity of MEPCM/GO-CNT is enhanced with the amount of the hybrid fillers increased, while the growth rate of the thermal conductivity is decreased. Because the content of CNTs increases as well, which leads to the improved interface thermal resistance. With a hybrid filler loading of 0.6 wt%, the thermal conductivity of the microcapsule reaches to 0.3821 W/mK, 2.95 times of the microcapsules without any filler. The thermal conductivity enhancement mechanism is proposed in Fig. 8. CNTs act as the heat conductive bridge between adjacent GO sheets and contribute to the formation of 3D network structure as thermal conductive pathways. Moreover, the 0D contact geometry of individual GO or CNT is replaced by a 1D linear contact of GO and CNT hybrid fillers, resulting in improved contact areas and decreased thermal interface resistance [27]. In a word, the combination of GO and CNTs outperforms single GO or CNTs in the enhancement of thermal conductivity of microcapsules, and the thermal energy charging/discharging rate of microcapsule with GO-CNT hybrid filler is largely enhanced.

3.4. Thermal storage properties of the microcapsules

Fig. 9 presented the latent heat of the microcapsules with different fillers. In our previous work [55], the average latent heat of the microcapsules with individual GO was 170 J/g and had nothing to do with the addition of GO. In the present work, we studied the effect of the

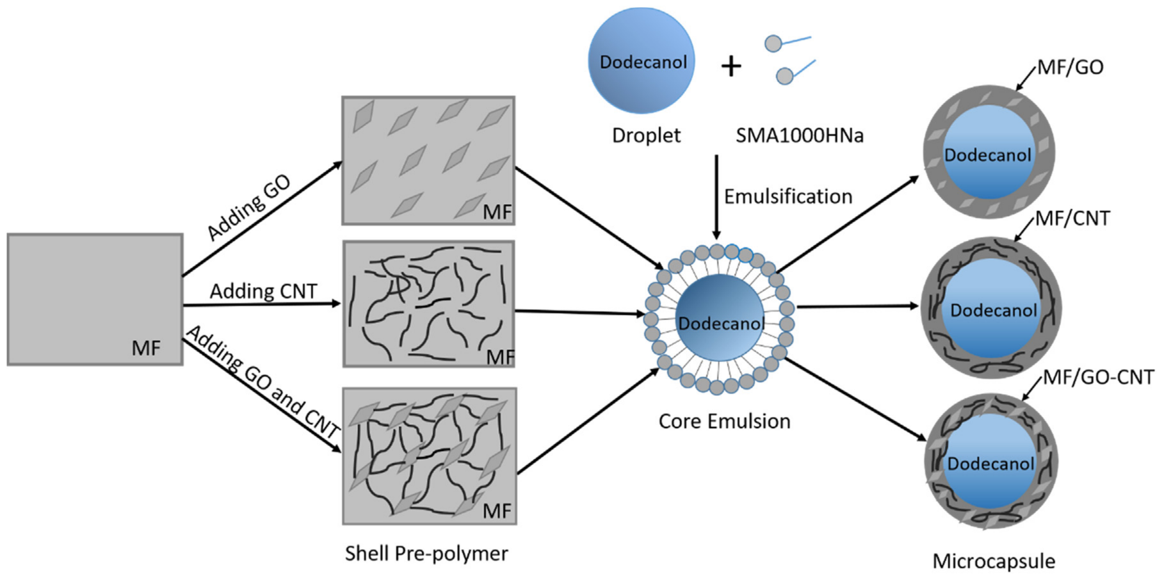

Fig. 8. Schematic diagrams of MEPCM with GO or CNT.

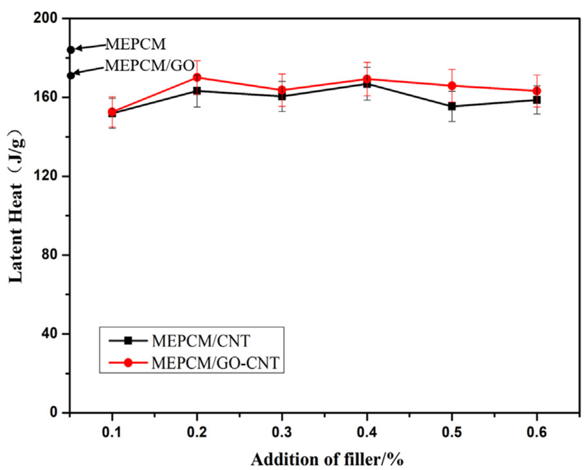

Fig. 9. Latent heat of the microcapsules with CNT or GO-CNT.

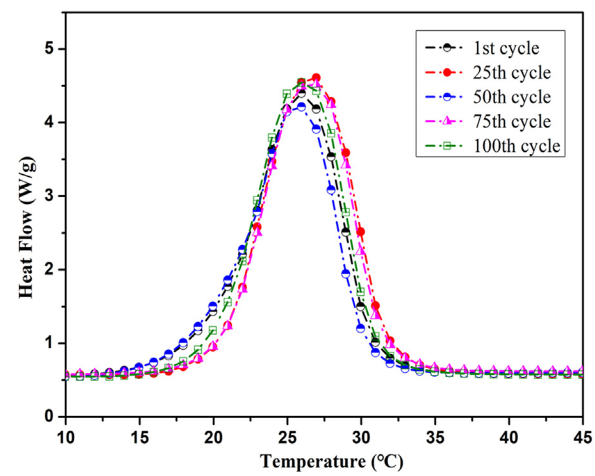

Fig. 10. DSC curves of MEPCM/GO-CNT after several thermal cycles.

addition of CNTs or the combination of GO and CNT on the latent heat of the microcapsules. With 0–0.6 wt% CNTs, the mean latent heat of the microcapsules was 158.8 J/g. The average latent heat of MEPCM/GO-CNT with 0–0.6 wt% GO-CNTs (3:1) hybrid filler was 162.9 J/g, slightly higher than the microcapsules with single CNT because of the poor dispersion of CNTs in the shell. As discussed in Fig. 2, the morphology of the microcapsules with CNT or hybrid filler was similar to that of the microcapsule without any filler, which indicated that the presence of fillers did not affect the encapsulation of the core material. In addition, the amount of the fillers was small and both of GO and CNT had a low density, thus the fillers occupied little of the whole quality. It indicates that the latent heat of the microcapsules just has a slight decrease with the addition of GO or CNT fillers, thus the energy storage capacity of MEPCM/GO-CNT still keeps at a high level.

3.5. Thermal reliability and stability of the microcapsules

Thermal cycle reliability and thermal stability are important characteristics of microcapsules for thermal energy storage application. Fig. 10 presents the thermal cycle test results of the microcapsules with 0.6 wt% GO-CNT (3:1) hybrid filler after 1–100 heating-freezing cycles. The DSC curves of MEPCM/GO-CNT after 25, 50, 75 and 100 phase change cycles nearly coincide with that of MEPCM/GO-CNT after 1st phase change. There is no significant change with the latent heat and phase change temperature, which are 26.3 °C and 156.7 J/g after 100 thermal cycles. The latent heat only decreases by 3.8% after undergoing 100 thermal cycles. This indicates that the obtained microcapsule can keep its phase change behavior after 100 thermal cycles and it has good thermal cycle stability.

Fig. 11 shows the TG thermograms and corresponding derivative of mass change of MEPCM with GO or CNT. The weight loss curves of all the microcapsules show a two-step weight loss. The first stage is related with the evaporation of n-dodecanol between the temperature range of 50–250 °C, and the second stage is due to the degradation of melamine resin shell during the temperature range of 250–500 °C. MEPCM without any filler starts to degrade at about 89.7 °C (Fig. 11A) and undergoes the most rapid mass loss at 152.6 °C (Fig. 11B). Its weight loss percent is 82.1% between 50 and 250 °C, which is roughly consistent with the encapsulation ratio of n-dodecanol in MEPCM. Accordingly, the onset degradation temperatures and the temperatures of the peak weight loss rate of MEPCM/CNT, MEPCM/GO and MEPCM/GO-CNT are 107 °C, 115 °C, 142.4 °C, and 187 °C, 199.5 °C, 209.7 °C with the weight loss of 74.55%, 75.65%, 75.2%, respectively. The results suggest that both the onset temperature and the temperature of peak weight loss rate of MEPCM with GO or CNT are higher than those of MEPCM without any filler. Particularly, MEPCM with GO-CNT

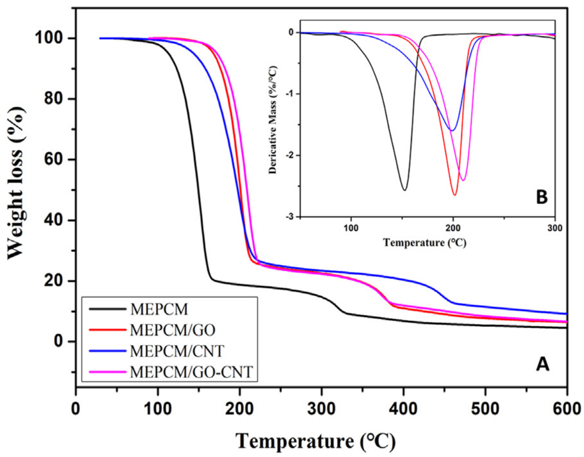

Fig. 11. (A) TG thermograms and (B) derivative of mass change of MEPCM with GO or CNT.

hybrid filler has the highest onset temperature and peak temperature of maximum weight loss rate. This demonstrates that the presence of GO or CNT in MEPCM increases the thermal stability of MEPCM and the combination of GO and CNT largely improves the onset degradation temperature due to the more compact shell with GO and CNT and the barrier property of GO and CNT. On the other hand, the slight difference of the weight loss for MEPCM and MEPCM with GO and CNT further indicates that the addition of GO and CNT has little effect on the thermal storage property of the microcapsules.

3.6. Photo-thermal conversion property of the microencapsulated phase change slurries

High thermal conductivity and good photo-thermal conversion property of the working fluids are desirable for direct absorption solar collectors [53], so the microcapsules with high thermal conductivity may have a potential application in DASC. The microcapsules modified by GO or CNT were respectively dispersed into deionized water with the aid of SDS to obtain stable suspensions, and their photographs were shown in Fig. 12(a). The suspension containing MEPCM with GO or CNT exhibited darker color than that with MEPCM. To characterize the stability of the suspensions, their zeta potentials were measured, as shown in Fig. 12(b). Generally, when the absolute zeta potential of the suspension was above 30 mV, it was physically stable. The zeta potential values of the four suspensions were all larger than 30 mV and were still above 30 mV after heating to 70 °C. This indicates that these suspensions have excellent stability and are suitable as the working fluid in DASCs.

Fig. 12(c) was the transmittance spectra of the suspensions. The transmittance T(λ) of the slurries with 0.01 wt% microcapsules in the wavelength range of 200–800 nm decreased to 62–72%. The slurry containing the microcapsule without any filler had the highest transmittance T(λ) of 72%, while the slurry containing MEPCM/GO-CNT had the lowest transmittance T(λ) of 62%. This may be due to the fact that the carbon nanomaterials such as GO and CNT had conjugated structure contributed to light absorption property, and the combination of GO and CNT can get a synergistic light absorption effect to further increase the light absorption caused by the light scatted or diffused at the interfaces. When the mass fraction of the microcapsules increased to

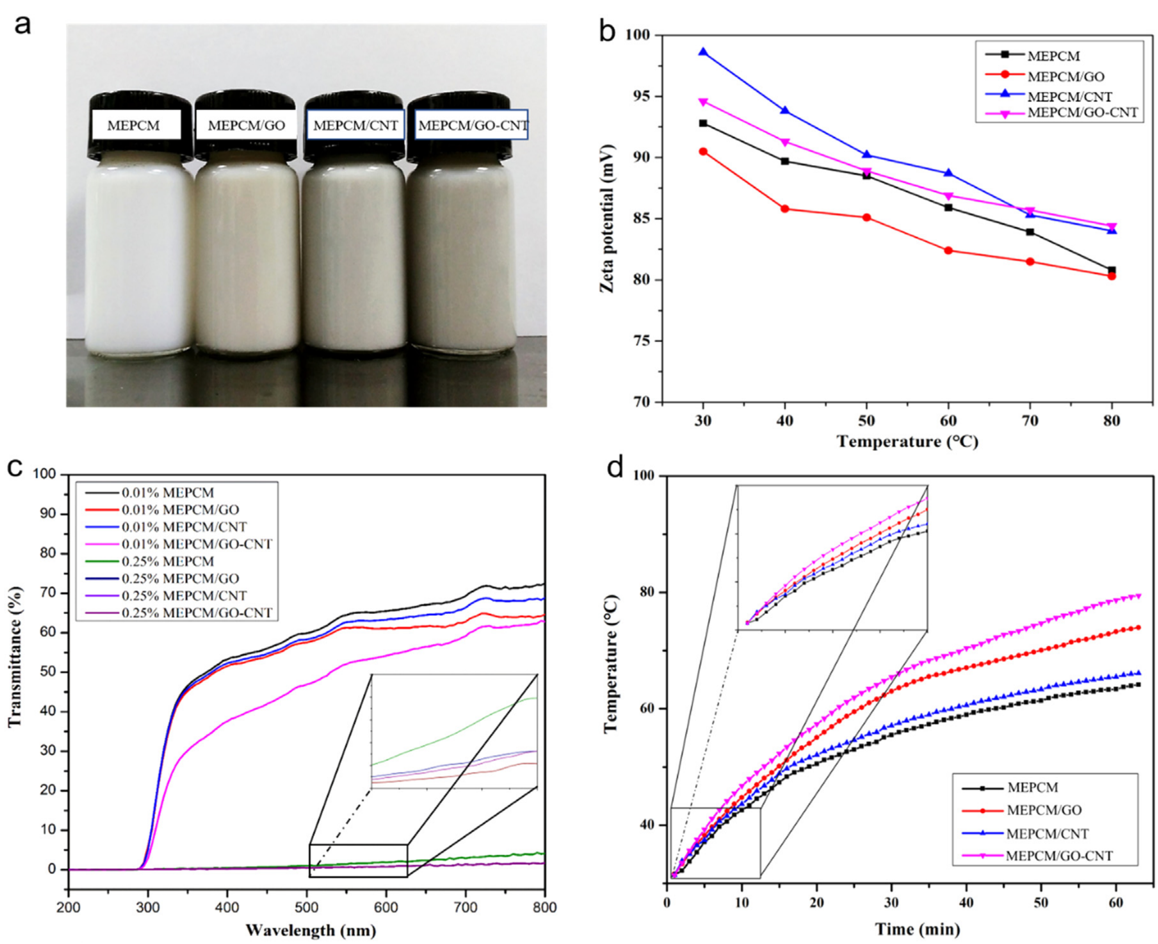

Fig. 12. Photographs of the suspensions (a). Zeta potentials of the suspensions as a function of temperature (b). Transmittance spectra of the suspensions (c). Variation in temperature with irradiation time of the suspensions (d).

0.25 wt%, the transmittance T(λ) of the slurries sharply decreased to below 5%. This indicated that the microencapsulated phase change materials slurries had good light absorption properties, which was a key property for DASCs.

Fig. 12(d) illustrated the temperature variations of the suspensions under irradiation. During the same irradiation time, the temperature of the suspension with MEPCM only increased to 64 °C, while the slurry containing MEPCM/GO-CNT had the highest temperature of 79 °C, indicating that the MEPCM/GO-CNT dispersed slurry had better photo-thermal conversion performance. Compared with MEPCM, MEPCM/GO-CNT possessed higher thermal conductivity, darker color and better light absorption property, which contributed to the better photo-thermal conversion performance of MEPCM/GO-CNT dispersed slurry and made it a potential working fluid for DASCs.

4. Conclusion

The microcapsules with dodecanol core and melamine-formaldehyde (MF) resin shell modified by graphene oxide (GO) and carbon nanotube (CNT) hybrid filler were prepared via in-situ polymerization. Studies on the morphology of the microcapsules revealed that the addition of GO and CNT had little influence on the spherical structure of the microcapsules. Studies on the phase change properties exhibited that the latent heat of the microcapsules was slightly decreased with the employ of GO and CNT. Compared with the microcapsule with only GO or CNT, the thermal conductivity of MEPCM/GO-CNT was largely enhanced by 195% at most. Furthermore, the obtained microcapsule had good thermal reliability and thermal stability. TGA results indicated that the onset degradation temperature and peak weight loss rate temperature of MEPCM/GO-CNT were increased by 52.7 °C and 57.1 °C, respectively. The obtained microcapsules were dispersed into deionized water to obtain stable slurries, and the suspension containing MEPCM/GO-CNT had the best photo-thermal conversion performance. The high thermal conductivity, good optical absorption property, and photo-thermal conversion performance made the microcapsule dispersed slurry a potential working fluid for DASCs.

Acknowledgements

This work was financially supported by the National Natural Science Foundation of China (21676096 and 21406073) and Guangzhou Jointas Chemical Ltd. Co. (China).

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